Energy-Based Test-Time Adaptation for Object Detection:

Towards Efficient Detection under Large Distribution Shifts

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Abstract

Object detection models exhibit significant performance degradation when deployed in environments that differ from their training distribution. Distribution shifts arising from weather conditions, illumination changes, or sensor variations can reduce detection accuracy substantially, rendering pretrained models unreliable for safety-critical applications. We propose an energy-based test-time adaptation (ETA) framework for object detection that enables pretrained models to adapt during inference under strict operational constraints: no access to source training data, single-pass processing of target samples, and fully unsupervised adaptation without ground-truth labels. Our method trains an energy model to quantify detection reliability by distinguishing in-distribution predictions from out-of-distribution failures. During preparation, we use synthetic corruptions applied to clean training images to generate degraded detections, training the energy model to assign low energy to accurate predictions and high energy to errors. The energy function operates on region-level features extracted from the detector's Feature Pyramid Network, producing spatial energy maps that localize prediction failures. At test time, we perform lightweight adaptation by updating only the batch normalization layers. We evaluate ETA on the Cityscapes-C benchmark across five corruption types relevant to autonomous driving: fog, motion blur, snow, brightness, and defocus. With a single adaptation iteration, ETA reaches a mean mAP of 24.1, outperforming the single-iteration baseline of the SOTA test-time adaptation method AMROD (22.4 mAP) and substantially improving over the unadapted source model (16.4 mAP). The gains are especially pronounced under severe corruptions such as snow, where ETA achieves 15.3 mAP compared to AMROD's 8.0 mAP, underscoring its robustness under extreme distribution shifts. Together, these results indicate that energy-based adaptation offers a computationally efficient and practically viable strategy for robust object detection in dynamic, real-world environments.

1 Introduction

Object detection models are fundamental to safety-critical applications including autonomous driving, robotic navigation, and surveillance systems. However, these models suffer from severe performance degradation when deployed in environments that differ from their training distribution. Distribution shifts arising from variations in weather conditions, illumination, sensor characteristics, or scene composition can reduce detection accuracy by over 50% [1], rendering pretrained models unreliable for real-world deployment. Consider an autonomous vehicle trained on clear-weather driving data: when encountering fog or snow for the first time, its object detector may fail to recognize pedestrians or vehicles, creating dangerous situations. Traditional domain adaptation methods could address these failures, but they impose impractical requirements—access to target domain data during training, multiple passes through adaptation data, and often ground-truth annotations [2, 3]. These constraints are incompatible with real-world deployment scenarios where models must adapt continuously to novel, unpredictable environments without human intervention. Can we enable pretrained object detection models to adapt reliably to novel target domains at test time, using only unlabeled target data observed in a single pass, without access to the original training data?

Test-time adaptation (TTA) has emerged as a promising paradigm that operates under precisely these constraints [4, 5]. Unlike traditional domain adaptation, TTA methods update model parameters during inference without source data access, process target samples sequentially without revisitation, and adapt in a fully unsupervised manner. Recent TTA approaches have achieved success in image classification through entropy minimization [5], batch normalization updates [6], and contrastive learning [7]. However, most TTA research focuses on classification or semantic segmentation; limited work addresses object detection [1], which presents unique challenges due to its structured outputs combining localization, objectness, and classification.

A parallel line of research has demonstrated the effectiveness of energy-based models (EBMs) for quantifying prediction reliability. Energy-based test-time adaptation (ETA) was recently introduced for depth completion [8], where an energy function learned to distinguish reliable predictions (low energy) from unreliable ones (high energy). By minimizing predicted energy at test time, the depth completion model successfully adapted to novel target domains. This approach is particularly appealing for object detection because it provides a principled mechanism to evaluate the complex, multi-modal predictions produced by detectors.

Inspired by the success of ETA in depth completion, we adapt this framework to object detection. Our key technical contributions are:

1. **Energy-based reliability quantification for object detection**: We design an energy model that evaluates detection reliability through learned feature representa-

tions extracted from a pretrained detector's Feature Pyramid Network, producing spatial energy maps that localize prediction errors.

2. **Lightweight test-time adaptation**: We implement an efficient adaptation strategy that updates a small subset of model parameters, maintaining real-time inference speeds.

Object detection forms a critical component of computer vision systems deployed in autonomous driving, robotics, and surveillance applications. However, models trained on curated datasets often experience substantial performance degradation when encountering distribution shifts arising from variations in illumination, weather conditions, sensor characteristics, or scene composition [5, 4]. While traditional domain adaptation methods can mitigate these effects, they impose impractical requirements for real-world deployment: access to target domain data during training, multiple passes through adaptation data, and often ground-truth annotations [2, 3].

Test-time adaptation (TTA) addresses these limitations by enabling model updates during inference under strict operational constraints [4]. TTA methods operate without access to original training data, process target samples in a single pass as they arrive sequentially, and adapt in an unsupervised manner without ground-truth labels. These properties make TTA particularly suitable for systems that must continuously operate in changing environments where retraining is infeasible.

1.2 Approach and Contributions

Recent work has demonstrated the effectiveness of energy-based models (EBMs) for test-time adaptation in depth completion, where an energy function quantifies prediction reliability relative to the source distribution [8]. Inspired by this success, we adapt the energy-based test-time adaptation framework to object detection, addressing unique challenges in this domain. There are a variety of technical challenges particular to this problem. First, object detection produces heterogeneous outputs (bounding boxes, objectness scores, class distributions) unlike the homogeneous depth maps in prior work. Furthermore, detections are localized to object regions rather than dense pixel-wise prediction. Finally, detection requires matching predictions to ground truth via Intersection-over-Union (IoU) thresholds

In light of these challenges, we provide the following contributions:

- We adapt the energy-based test-time adaptation framework from depth completion to object detection, designing an energy model that evaluates detection reliability through learned feature representations. This is a lightweight adaptation strategy that updates a small subset of parameters while maintaining real-time inference.

- We provide extensive preliminary experimental validation, showing that ETA is competitive with other test-time adaptation methods especially in scenarios where compute is limited or iteration speed is crucial.

2 Background

2.1 Object Detection

Object detection is the task of identifying and localizing objects within an image. Given an input image, a detector must predict both where objects are (through bounding boxes) and what they are (through class labels). Modern object detection architectures follow two main paradigms: two-stage detectors like Faster R-CNN [9] first generate candidate object regions and then classify them, while one-stage detectors like YOLO [10] and RetinaNet [11] directly predict bounding boxes and classes in a single forward pass. Two-stage detectors generally achieve higher accuracy at the cost of slower inference, making them well-suited for applications where detection quality is paramount.

Formally, an object detector G_θ parameterized by θ takes an input image $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$ and produces a set of detections $G_\theta(\mathbf{I}) = \{(\mathbf{b}_i, o_i, \mathbf{p}_i)\}_{i=1}^{N}$, where each detection i consists of a bounding box $\mathbf{b}_i = (x, y, w, h) \in \mathbb{R}^4$ specifying its spatial location and size, an objectness score $o_i \in [0, 1]$ indicating confidence that the region contains an object, and a categorical distribution $\mathbf{p}_i \in \Delta^{C-1}$ over C object classes. This work employs Faster R-CNN implemented in Detectron2 [12], a modular PyTorch-based detection framework. Our baseline detector uses a ResNet-50 [13] backbone with Feature Pyramid Network (FPN) [14] for multi-scale feature extraction, a Region Proposal Network (RPN) to generate candidate object regions, and ROI heads that refine proposals and predict final class labels. This architecture has demonstrated robust performance across diverse datasets and serves as a strong foundation for studying test-time adaptation.

2.2 Energy-Based Models

Energy-based models (EBMs) provide an alternative framework for probabilistic modeling that defines distributions implicitly through an energy function rather than through explicit density estimation [15]. The key insight is to assign a scalar energy value to each input, where low energy corresponds to high probability (in-distribution samples) and high energy indicates low probability (out-of-distribution or anomalous inputs). This approach offers significant flexibility in modeling complex distributions without requiring the partition function to be tractable during training or inference.

Mathematically, an EBM defines a probability distribution via the Boltzmann distribution: $p_\phi(\mathbf{x}) = \frac{\exp(-E_\phi(\mathbf{x}))}{Z(\phi)}$, where $E_\phi : \mathcal{X} \to \mathbb{R}$ is the energy function parameterized by ϕ and $Z(\phi) = \int_{\mathcal{X}} \exp(-E_\phi(\mathbf{x}))\, d\mathbf{x}$ is the partition function that normalizes the distribution. While computing $Z(\phi)$ is generally intractable, many applications only require relative energies between samples rather than absolute probabilities, enabling practical use with-

out explicit normalization. For object detection, we depart from modeling input images directly and instead define an energy function over detection outputs: $E_\phi : G_\theta(\mathbf{I}) \to [0, 1]$. This design choice is motivated by recent observations that detection features provide a more stable and domain-invariant representation than raw image features, which can vary dramatically across different visual conditions [16]. By learning to assign low energy to detections that resemble the source training distribution and high energy to unreliable or corrupted detections, the energy model serves as a learned quality metric that can guide test-time adaptation without requiring ground-truth annotations.

2.3 Test-Time Adaptation

Test-time adaptation (TTA) addresses a critical challenge in deploying machine learning models: how to maintain performance when test data differs from the training distribution. Unlike traditional domain adaptation, which assumes access to source training data and may allow multiple passes over target data, test-time adaptation operates under strict real-world constraints. First, the original training data is unavailable at test time, which is common in deployment scenarios due to storage limitations, privacy concerns, or proprietary restrictions. Second, target data arrives sequentially in a single-pass setting—the model must adapt to each sample as it arrives without the ability to revisit previous examples. Third, no ground-truth labels are provided for the target domain, requiring the adaptation mechanism to be fully unsupervised. Finally, adaptation must maintain real-time inference speeds, precluding expensive optimization procedures.

2.4 Batch Normalization

Batch normalization (BN) [17] is a widely-used technique that normalizes intermediate activations in neural networks to accelerate training and improve generalization. For each feature channel, BN computes the mean μ and variance σ^2 across a mini-batch during training, then normalizes activations to have zero mean and unit variance. Two learnable affine parameters—scale γ and shift β—are applied after normalization to restore the network's representational capacity. During inference, BN typically uses running statistics (exponential moving averages of μ and σ^2 accumulated during training) rather than batch statistics to ensure consistent predictions. However, this reliance on training-time statistics can be problematic when test data differs significantly from the training distribution, as the normalization may no longer be appropriate for the shifted input domain.

3 Methodology

Our energy-based test-time adaptation framework consists of two components operating in concert (Figure 3.1).

1. **Object Detector** (G_θ): Pretrained Faster R-CNN that processes input images and produces detections

2. **Energy Model** (E_ϕ): Convolutional network trained offline to evaluate detection reliability; remains frozen during deployment

While the energy model is trained with supervision on the source domain during the preparation phase, the test-time adaptation procedure itself is fully unsupervised, requiring no ground-truth annotations for target domain data.

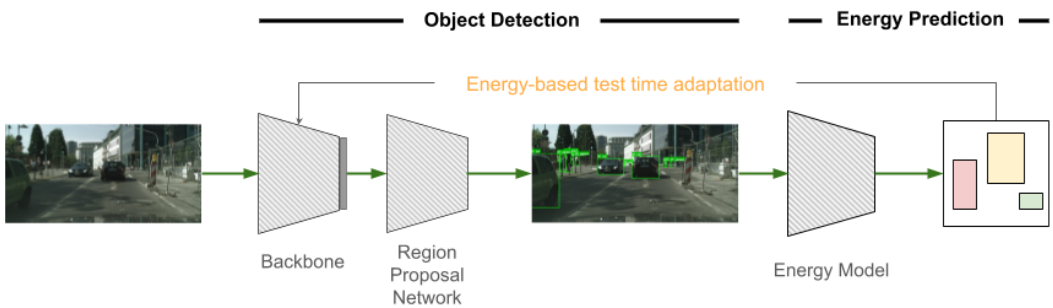

Figure 3.1: Diagram of energy test adaptation framework for depth completion. Striped boxes indicate layers frozen at test time, gray indicates trainable.

3.1 Energy-Based Model Training

Using a similar setup to the ETA framework for depth completion [8], we train the energy model using a discriminative objective that distinguishes in-distribution from out-of-distribution box detections. We model detection reliability using a region-level energy function E_ϕ that operates on RoI-aligned features extracted from Faster R-CNN with FPN (Feature Pyramid Network) backbone. Given an input image $\mathbf{I}$ and its detected bounding boxes $G_\theta(\mathbf{I})$, we first apply the detector backbone and multi-level RoIAlign to obtain a fixed-size feature tensor for each proposal:

$$\mathbf{f}_i = \text{RoIAlign}\big(\text{FPN}(\mathbf{I}), , \mathbf{b}_i\big), \quad \mathbf{f}_i \in \mathbb{R}^{C\times H\times W}, \tag{3.1}$$

where $\mathbf{b}i$ denotes the i-th predicted box, and the pooler aggregates features across all FPN levels using the same configuration as the detector's RoI heads (7×7 output, ROIAlignV2, multi-scale pooling). These per-region features are then passed to a dedicated convolutional energy head $E\phi$:

$$\mathbf{e}i = E\phi(\mathbf{f}i) = \sigma\big(\text{Conv}^{(L)}\phi(\mathbf{f}i)\big), \quad \mathbf{e}i \in [0,1]^{1\times H'\times W'}, \tag{3.2}$$

where $\text{Conv}^{(L)}\phi$ is a deep CNN and σ is the sigmoid activation. Concretely, $E\phi$ consists of a sequence of 2D convolutions with progressively increasing channel width and spatial downsampling:

$$256 \xrightarrow[\text{5×5, stride 2}]{\text{Conv + LeakyReLU}} 64 \xrightarrow{\cdots} 128 \xrightarrow{\cdots} 256 \xrightarrow{\cdots} 256 \xrightarrow{\cdots} 512 \xrightarrow{\cdots} 512, \tag{3.3}$$

followed by a final 3×3 convolution mapping $512 \to 1$ channel and a sigmoid nonlinearity. Each intermediate layer uses a 5×5 kernel with stride 2 and LeakyReLU activation ($\alpha = 0.2$). This design yields a low-resolution spatial energy map $\mathbf{e}_i$ for each RoI, where higher values correspond to higher predicted detection error.

To train E_ϕ, we construct supervision from paired clean–corrupted Cityscapes images and ground-truth annotations. For each image, we run the frozen detector on both the clean and corrupted versions, then compute a per-detection error signal by comparing predicted boxes to ground-truth boxes via a combined localization and scale discrepancy measure. Specifically, for each predicted detection i, we first compute a scalar geometric error that measures how well it matches the closest ground-truth box. Let $\mathbf{c}_i^{\text{pred}} \in \mathbb{R}^2$ and $\mathbf{s}_i^{\text{pred}} \in \mathbb{R}^2$ denote the center and size (width, height) of the i-th predicted box, and $\mathbf{c}_j^{\text{gt}}, \mathbf{s}_j^{\text{gt}}$ the corresponding quantities for ground-truth box j. The per-detection error is

$$\Delta_i = \begin{cases} \min\limits_j \left(\left\| \mathbf{c}_i^{\text{pred}} - \mathbf{c}_j^{\text{gt}} \right\|_2 + \dfrac{1}{2} \left\| \mathbf{s}_i^{\text{pred}} - \mathbf{s}_j^{\text{gt}} \right\|_2 \right), & \text{if match exists,} \\ 1.0, & \text{otherwise,} \end{cases} \tag{3.4}$$

where the second case corresponds to images with only predictions or only ground truth, and we assign a large constant penalty to reflect complete detection failure. We convert this unbounded error into a bounded "target energy" via a Gibbs-like transform:

$$\tilde{e}_i = 1 - \exp\left(-\frac{\Delta_i}{T}\right), \tag{3.5}$$

where $T > 0$ is a temperature hyperparameter. This exponential mapping assigns near-zero energy to accurate predictions and approaches unity for large errors. Note that $T = 200$ was used empirically for all experiments in this paper.

The energy head E_ϕ produces a spatial energy map $\mathbf{e}_i \in [0,1]^{1\times H'\times W'}$ for each detected region i. We train the model with an ℓ_2 loss between the predicted map and the scalar target energy $\tilde{e}_i$ (broadcast spatially):

$$\mathcal{L}_{\text{energy}} = \frac{1}{N} \sum_{i=1}^{N} \|\mathbf{e}_i - \tilde{e}_i\|_2^2, \tag{3.6}$$

where N is the number of detections in the batch and the norm is computed over all spatial locations. The key innovation is the RoI-based approach that enables per-detection energy estimation, providing more targeted adaptation signals compared to global energy metrics or a per-bounding box confidence score.

3.2 Adapting Batch Normalization Layers

We perform test-time adaptation by updating only the batch normalization (BN) layers while keeping all other model parameters frozen. This approach enables efficient adaptation without requiring gradient computation through the entire network.

Detectron2 models use `FrozenBatchNorm2d` layers by default, which maintain fixed statistics during both training and inference. To enable test-time adaptation, we convert all `FrozenBatchNorm2d` layers to standard `BatchNorm2d` layers, preserving the learned affine parameters (scale γ and shift β) and running statistics (mean μ and variance σ^2). This conversion yields 53 BN layers distributed throughout the ResNet-50 FPN backbone and detection heads (RPN and ROI heads). Each BN layer contains two affine parameters—γ for scaling and β for shifting—which modulate the normalized activations. We make only these affine parameters trainable while freezing all convolutional weights, biases, and other network parameters, resulting in 53,120 trainable parameters total. This represents a tiny fraction of the full model (approximately 0.12% of the 44M total parameters), enabling efficient adaptation with minimal computational overhead.

3.3 Energy-Based Test-Time Adaptation

Following the energy-guided adaptation framework, we perform test-time adaptation by updating only the batch normalization (BN) layers while keeping all other model parameters frozen. For each test image, we perform gradient descent on the BN affine parameters to minimize the energy predicted by our trained energy model E_ϕ. We first run the detector on the corrupted image to generate proposals and detections with score threshold $\tau = 0.05$. For each detection above threshold, we extract ROI-aligned features $\mathbf{f}_i$ using the same multi-level FPN pooling as during energy model training. These features are passed through the frozen energy model to obtain per-detection energy maps, which we average across all detections to compute the adaptation loss:

$$\mathcal{L}_{\text{energy}} = \frac{1}{N} \sum_{i=1}^{N} E_\phi(\mathbf{f}_i), \tag{3.7}$$

where N is the number of detections. We backpropagate this loss through the backbone with BN layers in training mode, updating only the affine parameters γ and β via SGD with learning rate α and momentum 0.9. This process repeats for K iterations per image.

Critically, we set the detector to evaluation mode while keeping BN layers in training mode. This allows BN statistics to accumulate from the current image while preventing dropout or other stochastic behavior. We use momentum-based SGD rather than Adam

to maintain stability during the brief adaptation period. To prevent catastrophic forgetting across the test sequence, we reset BN parameters to their initial state periodically (every image in our experiment), balancing adaptation to local corruption patterns with retention of the pre-trained feature statistics. The energy model remains frozen throughout adaptation, serving purely as a learned signal of detection quality that guides parameter updates toward more reliable predictions on corrupted inputs.

4 Results

4.1 Experimental Setup

In this work, all experiments are conducted using a single object detector backbone, namely Faster R-CNN [9] with a ResNet50 feature extractor. This choice provides a strong baseline for object detection while remaining computationally tractable for repeated test-time adaptation experiments. The models are initialized with Detectron2 weights and configuration files provided in the AMROD paper, with pre-training performed on the standard Cityscapes dataset [18]. Using a pre-trained detector allows us to focus our evaluation on the effects of energy-based test-time adaptation rather than on training a detector from scratch, ensuring that performance differences are attributable primarily to adaptation strategies.

Each energy model is trained on a subset of the Cityscapes *train* split, consisting of 860 images from the cities of Düsseldorf, Hamburg, Ulm, and Zurich. These images are synthetically corrupted with the maximum severity level (5) of the respective corruption type, allowing the energy models to learn to predict reliable adaptation signals under extreme conditions. The Cityscapes-C dataset [19] serves as the primary benchmark for evaluating model robustness. Cityscapes-C was originally designed to assess the impact of 15 distinct types of image corruption, each applied at five levels of severity, ranging from mild to extreme. In our experiments, we generate a custom evaluation set by applying these corruptions at the highest severity level to the clean Cityscapes validation images, treating each corruption as a separate target domain comprising 500 images. This setup allows us to systematically investigate the generalization and robustness of the detector under diverse, challenging conditions.

For the TTA task, we prioritize a subset of five corruption types most relevant to autonomous driving applications: **Fog, Motion, Snow, Brightness, and Defocus**, following prior work in [1]. During short-term adaptation, the model's adaptation state is reset between images to ensure independence of adaptation results and to prevent accumulation of biases across images.

Throughout all experiments, the primary evaluation metric is mAP@0.5, or mean average precision at an inter IoU (intersection over union) threshold of 0.5. This metric measures both the precision and recall of the detector across all target domains and provides a standard benchmark for object detection performance. It allows us to quantitatively assess the improvements achieved through test-time adaptation under a wide variety of corruption types, highlighting the robustness and reliability of the proposed energy-based approach.

4.2 Comparison with Baseline Methods

We compare ETA-Object with seven source-free baselines across various settings, including Source [20], Tent [5], IRG [21], MemCLR [22], CoTTA [23], SVDP [24], and AMROD [1]. Specifically, "Source" represents the source model without adaptation. Tent [5] updates the affine parameters through entropy minimization in TTA. Furthermore, MemCLR and IRG are SFDA object detection methods [22]. CoTTA [23] utilizes a fixed threshold for pseudo-labeling and random neuron recovery to tackle TTA, while SVDP explores sparse visual prompts for TTA dense prediction.

Table 4.1: Experimental results (mAP@0.5) for 5 rounds of Cityscapes-to-Cityscapes-C TTA task. Bold is best, italics is second best.

Condition	Fog	Motion	Snow	Brightness	Defocus	Mean	Gain
Source	36.1	8.1	0.2	31.0	6.7	16.4	/
Tent	30.7	10.2	0.8	27.9	12.4	16.4	0.0
COTTA	40.8	10.4	0.6	36.5	9.2	19.5	+3.1
SVDP	45.0	15.4	3.8	47.8	19.9	26.4	+10.0
IRG	45.6	17.7	6.0	46.7	21.9	27.6	+11.2
MemCLR	45.0	*18.1*	5.1	46.8	22.4	27.5	+11.1
AMROD	49.4	**18.5**	*8.0*	**52.9**	24.7	**30.7**	**+14.3**
ETA-OD (Ours)	34.0	16.7	**15.3**	*47.9*	10.4	24.9	+8.5

4.3 Iteration Ablation Study

The number of test-time adaptation iterations presents a trade-off between computational cost and performance (mAP@0.5). A key advantage of the Energy-Based Test-time Adaptation (ETA) framework is its support for multiple iterative updates per test sample. Unlike entropy-based methods that rely on a single forward pass, ETA can perform several iterations independently, offering flexibility to adjust the number of iterations according to the application.

For real-time scenarios, such as autonomous driving, fewer iterations strike a balance between accuracy and speed, maintaining high FPS while still achieving notable performance gains. Conversely, for offline tasks where latency is less critical, additional iterations can be employed to maximize accuracy. Compared to AMROD, ETA achieves competitive or better results even with a single iteration, making it both lightweight and computationally efficient.

From Table 4.2, several trends emerge. First, a single iteration of ETA already outperforms AMROD in terms of mean mAP, suggesting efficiency in both computation and accuracy. Interestingly, while increasing the number of ETA iterations improves performance on some corruption types (e.g., Motion and Snow), it leads to a decrease in Fog

Table 4.2: Comparison of mAP@0.5 across different corruptions on Cityscapes-C. Bold values highlight where ETA-OD (Ours) outperforms AMROD with a single iteration.

Method	Rounds	Fog	Motion	Snow	Brightness	Defocus	Mean
Source	0	36.1	8.1	0.2	31.0	6.7	16.4
Single Iteration							
AMROD	1	39.0	10.4	0.8	**48.0**	13.8	22.4
ETA-OD (Ours)	1	**43.5**	**14.8**	**1.2**	47.5	13.7	**24.1**
Multiple Iterations							
AMROD	5	**49.4**	**18.5**	8.0	**52.9**	**24.7**	**30.7**
ETA-OD (Ours)	5	34.0	16.7	**15.3**	47.9	10.4	24.9

and Defocus performance, highlighting potential sensitivity to certain conditions. AMROD, in contrast, tends to benefit more consistently from additional iterations, particularly in Fog and Defocus, achieving the highest scores in those categories. Additionally, ETA maintains strong performance in Brightness and shows notable gains over the source model even with a single iteration. These patterns underscore the flexibility of ETA, where the number of iterations can be tailored to application requirements, trading off between computational cost and robustness across corruption types.

4.4 Qualitative Results

Figure 4.1 shows qualitative comparisons on motion blur (top) and snow (bottom) corruptions. Without adaptation (a), the source model produces missing detections, poor localization, and false positives. After applying ETA-OD (b), we observe: (1) recovery of missed detections, particularly for distant objects; (2) improved bounding box localization and tighter fits; (3) elimination of false positives; and (4) higher confidence scores for true detections. The improvements are especially pronounced under severe corruption (snow), where ETA-OD detects multiple vehicles and persons completely missed by the source model, demonstrating effective adaptation to extreme distribution shifts.

(a) Source (b) ETA-OD (Ours)

Figure 4.1: Qualitative results on Cityscapes-C validation set with motion blur (top) and snow (bottom) corruptions. (a) Source model trained on clean images fails to detect distant vehicles and produces false positives under distribution shift. (b) Our ETA-OD method recovers missed detections, improves localization, and increases detection confidence through energy-guided batch normalization adaptation.

5 Related Work

Our work builds upon three main areas of research: test-time adaptation, domain adaptation for object detection, and energy-based models. We discuss the most relevant prior work and clarify how our approach differs.

5.1 Test-Time Adaptation

Test-time adaptation addresses distribution shift under strict operational constraints: no access to source training data, single-pass processing of target data, and unsupervised adaptation without ground-truth labels [4].

Early TTA methods employed auxiliary self-supervised tasks to guide adaptation. Sun et al. [25] proposed training models on rotation prediction as a pretext task, using this signal to update model parameters at test time when encountering distribution shifts. Liu et al. [26] extended this with jigsaw puzzle solving and investigated when self-supervised TTA succeeds or fails. While effective for classification, these methods require carefully designed auxiliary tasks that may not transfer well to structured prediction tasks like object detection.

Entropy-based adaptation has emerged as a more direct approach. TENT [5] updates batch normalization statistics by minimizing prediction entropy, encouraging confident outputs on target domain data. This simple yet effective strategy has been widely adopted but can suffer from confirmation bias when initial predictions are incorrect. Wang et al. [23] extended TENT to continual adaptation scenarios with CoTTA, using a teacher-student framework and stochastic restoration to prevent catastrophic forgetting over long sequences. CoTTA represents the current state-of-the-art for continual TTA but was primarily designed for classification tasks.

Batch normalization adaptation exploits the observation that BN statistics capture domain-specific information. Schneider et al. [6] proposed adapting only BN layers while keeping all other parameters frozen, achieving robustness against common corruptions. This lightweight approach is computationally efficient but may have limited capacity to handle severe distribution shifts.

Contrastive TTA leverages representation learning. Chen et al. [7] proposed using contrastive objectives to align target features with source representations stored in memory. While promising, this requires maintaining feature banks which can be memory-intensive for deployment.

Comparison to our work: Unlike prior TTA methods that rely on entropy minimization or self-supervised signals, we introduce an energy-based objective that explicitly quantifies prediction reliability. Our energy model is trained offline to recognize patterns

indicative of distribution shift, providing more principled guidance than entropy alone. Furthermore, most TTA work focuses on classification [5, 6] or semantic segmentation [7]; little research addresses object detection [1] with limited computational resources, which we specifically target.

5.2 Domain Adaptation for Object Detection

Domain adaptation for object detection has been extensively studied under less restrictive settings than TTA.

Supervised domain adaptation assumes access to both source and target data simultaneously. Chen et al. [20] proposed Domain Adaptive Faster R-CNN, using adversarial training to align image-level and instance-level features across domains. Saito et al. [27] introduced strong-weak distribution alignment, leveraging both global and local feature alignment. While effective, these methods require simultaneous access to source and target data, which violates TTA constraints.

Source-free domain adaptation relaxes the requirement for source data access. Li et al. [28] proposed Model Adaptation (MA), which adapts models using only target data by leveraging information encoded in the pretrained model. Yang et al. [29] extended this to generalized source-free adaptation with pseudo-labeling strategies. However, these methods still require multiple passes through the target dataset for training, unlike our single-pass TTA setting.

Adversarial perturbations A fundamental challenge in training the energy model is the absence of actual target domain data during the preparation phase. Past research has used adversarial perturbations to explore the space of possible out-of-distribution samples. The perturbed images $\tilde{\mathbf{I}}_s$ induce degraded predictions $\tilde{\mathbf{d}}_s = G_\theta(\tilde{\mathbf{I}}_s)$, which serve as synthetic out-of-distribution samples. Adversarial perturbations push inputs toward statistical outliers in high-density regions of the data manifold [30], causing prediction errors that simulate diverse failure modes without requiring specific assumptions about the target distribution. This enables training a single energy model that generalizes across multiple potential target domains.

TTA for object detection remains underexplored. Cao et al. [1] recently proposed test-time adaptation specifically for object detection under distribution shifts, using consistency regularization and pseudo-labeling. Their work is the most closely related to ours but differs in several key aspects: they do not use energy-based models, rely primarily on pseudo-labels which can accumulate errors, and do not explicitly model prediction reliability.

Comparison to our work: We operate under stricter TTA constraints (single-pass, no source access) compared to traditional domain adaptation. Unlike source-free methods that require multiple epochs over target data, we adapt online as samples arrive. Compared to [1], we introduce energy-based reliability quantification rather than relying solely on pseudo-labels, and our approach explicitly localizes prediction errors through spatial energy maps.

5.3 Energy-Based Models for Adaptation

Energy-based models define probability distributions implicitly through energy functions [15], providing a flexible framework for modeling complex distributions.

EBMs in computer vision have been applied to various tasks. Du and Mordatch [31] demonstrated implicit generation with EBMs for image synthesis. Gustafsson et al. [32] proposed using EBMs for deep probabilistic regression, showing improved uncertainty estimation. Song et al. [33] developed efficient training methods for EBMs using sliced score matching. These works establish EBMs as powerful tools for density estimation but do not address test-time adaptation.

Energy-based test-time adaptation was recently introduced by Yuan et al. [34] for image classification. TEA transforms pretrained classifiers into energy-based models and adapts by minimizing energy at test time. However, TEA computes a single scalar energy per image and was designed for classification, not addressing the spatial structure and multi-modal outputs of object detection.

ETA for depth completion was proposed by Chung et al. [8], which is the most directly related work to ours. They train a patch-based energy model to evaluate dense depth predictions conditioned on sparse depth inputs, using adversarial perturbations to synthesize out-of-distribution samples. At test time, they minimize predicted energy to align depth predictions with the source distribution. Their work demonstrates the effectiveness of energy-based adaptation for dense prediction tasks.

Comparison to our work: We adapt the ETA framework from depth completion [8] to object detection, addressing fundamentally different challenges. Depth completion produces homogeneous, dense outputs (depth values at each pixel), while object detection produces heterogeneous, sparse outputs (bounding boxes, objectness scores, class distributions). We design a new energy model architecture that processes detection features from Feature Pyramid Networks rather than depth maps, develop a multi-component adaptation loss that handles heterogeneous predictions, and demonstrate effectiveness on detection-specific benchmarks. Compared to TEA [34], we produce spatial energy maps rather than scalar energies, enabling localized adaptation for structured predictions.

6 Conclusions

This work investigated energy-based test-time adaptation (ETA) as a practical framework for improving object detection robustness under distribution shift. We asked whether pre-trained detectors could adapt reliably to novel corrupted environments at test time using only unlabeled target data observed in a single pass, without access to source training data. Our results demonstrate that ETA offers a viable approach to this challenge, with particular strengths in computational efficiency and adaptation to severe corruptions. Our main findings reveal a fundamental trade-off between iteration count and adaptation quality. With a single iteration, ETA-OD achieves a mean mAP of 24.1 across five corruption types, outperforming AMROD's single-iteration performance of 22.4 and substantially improving over the source model baseline of 16.4. This demonstrates that energy-guided adaptation provides meaningful improvements even under strict computational constraints. However, extending to five iterations yields mixed results: while performance on challenging corruptions like Snow improves dramatically (from 1.2 to 15.3 mAP, surpassing AMROD's 8.0), other corruption types experience degradation, with mean performance declining to 24.9 compared to AMROD's 30.7. These patterns suggest that ETA excels in scenarios prioritizing either computational efficiency or adaptation to extreme distribution shifts. For real-time applications like autonomous driving where inference speed is critical, single-iteration ETA provides superior performance to competing methods while maintaining minimal overhead. For severe corruptions where other methods struggle—particularly Snow, where ETA achieves nearly double AMROD's performance—the energy-based framework demonstrates robust adaptation even when predictions are highly unreliable. This behavior likely reflects the energy model's training regime, which exposed it to maximum-severity corruptions and learned to recognize patterns of extreme failure. The contribution of this work lies in adapting the energy-based test-time adaptation framework from dense prediction tasks to the structured, heterogeneous outputs of object detection. We designed an energy model operating on ROI-aligned features from Feature Pyramid Networks, developed a lightweight adaptation strategy updating only 53,120 batch normalization parameters (0.12% of the full model), and demonstrated that this approach provides competitive performance under computational constraints while excelling on challenging distribution shifts. Our results establish that energy-guided adaptation offers a principled alternative to entropy minimization and pseudo-labeling, with distinct advantages in specific deployment scenarios. Looking forward, several open questions remain. The performance degradation at higher iteration counts suggests that the current adaptation procedure may benefit from improved regularization or hybrid loss functions combining energy minimization with complementary objectives. The corruption-specific sensitivity we observed indicates that adaptive strategies—adjusting iteration count or learning rate based on detected shift severity—could

yield more consistent improvements. Finally, extending this framework to continual adaptation scenarios and evaluating on real-world benchmarks beyond synthetic corruptions would clarify whether energy-based methods can maintain stability over long deployment horizons. By demonstrating both the promise and limitations of energy-guided test-time adaptation for object detection, this work establishes a foundation for future research toward robust, deployable perception systems.

7 Future Work

There are several promising directions to extend this work beyond its current scope. A natural next step is to refine the adaptation model's training procedure. The present setup focuses primarily on batch normalization adaptation with energy guidance, but optimizing the interaction between the energy term, normalization statistics, and network stability remains an open question. Exploring improved schedules, stronger regularization, or hybrid adaptation losses—such as combining entropy minimization with energy-based terms—may lead to more robust and reliable adaptation behavior. Also add a 1x1 convolutional layer, see how that would work. Once that is complete, we can convert the repo into a more general plug-and-play framework.

An important avenue for future investigation involves developing more sophisticated reset mechanisms for batch normalization statistics. Currently, ETA employs a naive reset strategy that reinitializes batch normalization parameters to their pretrained values after each image, ensuring independence between adaptations but potentially discarding useful domain-specific information. In contrast, AMROD maintains accumulated statistics across the entire test sequence without resetting, which may explain its more consistent performance gains with multiple iterations. This suggests that the optimal reset strategy likely lies between these two extremes. Future work could explore adaptive reset policies that selectively retain or discard statistics based on detected distribution shift severity, temporal consistency of the input stream, or energy-based confidence measures. For instance, resetting could be triggered only when the energy model detects a significant domain transition, or statistics could be maintained with exponential decay to balance adaptation to recent samples with retention of longer-term patterns. Additionally, investigating class-specific or layer-specific reset schedules—where different components of the network retain statistics for different durations—may provide finer-grained control over the adaptation process. Understanding when and how to reset model state represents a fundamental question in continual test-time adaptation, and energy-based signals may offer principled guidance for these decisions.

A broader and more systematic evaluation is also essential. While current experiments test performance under corruptions such as fog and brightness, future work should expand to more challenging weather or lighting conditions including rain, nighttime scenes, and glare. Beyond synthetic datasets, deploying the method on real-world benchmarks such as BDD100K or nuScenes is critical to understanding practical robustness and domain transfer performance. These datasets capture diverse environments and camera conditions that would better stress-test the adaptation protocols.

In addition, future work should explore covariate shift quantification. Because the energy model produces meaningful uncertainty estimates, its outputs may serve as a proxy for measuring distribution shift severity. This could unlock dynamic adaptation strate-

gies, such as adjusting the learning rate, the number of test-time updates, or the strength of the energy guidance signal based on the detected shift level. Developing principled methods for using energy statistics as shift indicators represents an exciting intersection of calibration and adaptation research.

Several more advanced extensions could further broaden the impact of this line of work. Energy model training could benefit from richer perturbation strategies—such as Projected Gradient Descent, Carlini–Wagner–style adversarial methods, or naturalistic augmentations—to encourage smoother and more informative energy landscapes. Architectural improvements also offer potential gains: using attention mechanisms or transformer-based energy models may yield superior reliability estimation and better calibration. On the detection side, multi-scale adaptation that extends beyond the backbone and into components such as the Region Proposal Network may enhance robustness by improving proposal generation under challenging conditions.

Finally, scaling this framework to a wider range of pretrained models and performing comprehensive evaluations across multiple benchmarks would help establish the generality of energy-guided test-time adaptation. By comparing performance across architectures, datasets, and corruption types, future work can more clearly identify the strengths and limitations of the method and lay the groundwork for robust, deployable adaptive perception systems.

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A Supplementary Material

A.1 Code

The preliminary code is released publicly in this GitHub repo: https://github.com/chessgoose/eta-object-detection.

A.2 Training Hyperparameters

The energy models used for ETA were trained with slightly different learning rates depending on the corruption type. Specifically, the learning rate for the energy model and the training learning rate during test-time adaptation were tuned for each corruption to ensure stable and effective adaptation. The batch size and number of epochs were held constant across all corruptions (Batch Size = 32, Epochs = 2). Table A.1 summarizes the learning rates used in our experiments.

Table A.1: Learning rates for energy model training and ETA for different corruption types. Batch size and number of epochs are constant across all experiments.

Corruption Type	Energy Model LR	ETA Training LR
Fog	0.001	1
Motion	0.0005	2
Snow	0.001	1
Brightness	0.0005	0.2
Defocus	0.002	0.05